Machine Learning Final Project
Contribution Report

Shahin Safarli, Gulnisa Abdurahmanli, Jeyhuna Sevdiyeva, Seljan Khasiyeva, Suleyman Allahverdiyev

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Project: Ensemble Methods: Boosting vs. Bagging. **Release target:** `v1.0-final` after clean-clone verification. **Team-size note:** This is a 5-member team; the team's course-staff approval/acceptance for the non-standard team size is handled in the external course communication/Moodle record.

This report records the work allocation for the submitted project. The distribution is balanced by implementation difficulty, code volume, experiment ownership, testing responsibility, and written/slide material. The final GitHub history should match these responsibilities and the electronically agreed contribution statement.

Member Name	Modules/Code Sections	Report Sections	Experiments	Slides	Other Tasks
Shahin Safarli	`src/boosting/adaboost.py`; `src/boosting/gradient_boosting.py`; boosting exports and ensemble tests. Implemented SAMME AdaBoost, staged predictions, probability scoring, SAMME.R path, and Gradient Boosting bonus.	AdaBoost method, boosting math, scaling interpretation, Gradient Boosting bonus, and noise-sensitivity discussion.	AdaBoost scaling from 1–200 rounds; boosting side of noise robustness; Gradient Boosting vs. AdaBoost comparison.	Boosting method, AdaBoost scaling, head-to-head interpretation, and final conclusion slides.	Validation of staged predictions, estimator weights/errors, SAMME.R behavior, probability outputs, and final four-dataset runner integration. Approximate historical commit share from the supplied Git history: 23%. Suggested prefix: `boosting:`.
Gulnisa Abdurahmanli	`src/experiments/datasets.py`; `src/unsupervised/dbscan.py`; `src/utils/preprocessing.py`; `src/metrics/evaluation.py`; shared experiment utilities. Implemented authoritative loaders, exact largest-remainder selection, mixed-type fitted preprocessing, exact KD-tree DBSCAN, oversampling, metrics, and bias-variance helper.	Dataset contract, KD-tree DBSCAN method, Adult preprocessing and imbalance treatment, metrics explanation, noise robustness, and Breast Cancer bias-variance discussion.	Full-data DBSCAN k-distance and ARI analysis; seven-class Covertype setup; label-noise experiment; Breast Cancer bootstrap bias-variance decomposition.	Dataset, noise robustness, bias-variance, and imbalance/AUC explanation slides.	Validation of exact dataset sizes, DBSCAN parity and 50,000-point behavior, Adult preprocessing, metrics, oversampling, label flipping, and bias-variance outputs. Approximate historical commit share from the supplied Git history: 26%. Suggested prefix: `dbscan-metrics:`.
Jeyhuna Sevdiyeva	`src/trees/decision_tree.py`; assignment-path wrappers in `src/trees/`. Implemented CART nodes, weighted impurity, entropy/Gini, positive-gain split rule, predictions, probabilities, feature importances, and stump specialization.	Decision Tree method, baseline sanity-check section, sklearn comparison explanation, and tree/stump limitations.	Baseline tree experiment; sklearn tree reference comparison; tree component of 5-fold head-to-head evaluation.	Implementation map, baseline sanity-check, and decision-tree defense slides.	Tree tests for simple splits, no-gain leaves, edge cases, weighted stumps, feature importances, and four-dataset sklearn-gap behavior. Approximate historical commit share from the supplied Git history: 14%. Suggested prefix: `tree:`.
Seljan Khasiyeva	`src/unsupervised/pca.py`; `src/unsupervised/kmeans.py`. Implemented PCA centering/covariance/eigendecomposition, explained-variance ratios, K-Means++ initialization, Lloyd updates, inertia, restarts, and cluster prediction.	PCA/K-Means method, scree and elbow interpretation, cluster-label alignment analysis, and t-SNE bonus context.	PCA scree plots; PCA projections; K-Means elbow curves; K-Means ARI; full-data t-SNE bonus visualization for MNIST2Class.	Unsupervised evidence slide and related conclusion points.	PCA and K-Means validation in unit tests; review of all four datasets' figure readability and report consistency. Approximate historical commit share from the supplied Git history: 14%. Suggested prefix: `pca-kmeans:`.
Suleyman Allahverdiyev	`src/bagging/random_forest.py`; forest exports and wrapper. Implemented bootstrap sampling, OOB indices/score, feature subsampling, deterministic per-tree seeds, hard majority voting, probability alignment, and feature importances.	Random Forest method, bagging/variance explanation, OOB discussion, forest scaling interpretation, and MNIST2Class/seven-class Covertype conclusions.	Random Forest estimator scaling; max-depth scaling; OOB-vs-test comparison; Random Forest side of head-to-head and noise experiments.	Random Forest scaling, head-to-head result, and high-dimensional/noisy-data conclusion slides.	Forest tests for OOB score, probability shape, hard majority voting, deterministic behavior, feature importances, training accuracy, and worker-count propagation. Approximate historical commit share from the supplied Git history: 23%. Suggested prefix: `forest:`.

Table 1: Contribution table using the columns required by the project statement.

AI Tool Disclosure

AI coding and writing assistants were used to scaffold, debug, and polish parts of the project. The team reviewed, tested, and edited the outputs; every member affirms that they can defend the code and written claims assigned to them. Substantial AI usage is also disclosed in the main report.

Electronic Agreement / Signatures

The typed agreements below record each member's final review and approval of this revised contribution report and contribution allocation on 18 July 2026. Each member confirms that the descriptions are accurate and that they can defend the work assigned to them.

Member	**Signature / Electronic Agreement**	**Date**	**Agree**
Shahin Safarli	Typed e-agreement: SS	2026-07-18	✓
Gulnisa Abdurahmanli	Typed e-agreement: GA	2026-07-18	✓
Jeyhuna Sevdiyeva	Typed e-agreement: JS	2026-07-18	✓
Seljan Khasiyeva	Typed e-agreement: SK	2026-07-18	✓
Suleyman Allahverdiyev	Typed e-agreement: SA	2026-07-18	✓